

Sharif University

Department of Electrical Engineering

Final Project

Attention-Aware Mixed-Precision Quantization of Transformer Models

Course: Deep Learning

Semester 1404-1405

Project Proposal
Deep Learning Course

1 Background

Transformer models have become the dominant architecture in modern deep learning. Despite their remarkable performance, their deployment on edge devices remains challenging due to their large memory footprint and computational requirements.

Quantization is one of the most successful approaches for reducing model size and inference latency. Most existing post-training quantization (PTQ) methods formulate quantization as a layer-wise approximation problem.

Given an input activation X and a weight matrix W , they seek a quantized weight matrix $\hat{W}$ by minimizing

$$\min_{\hat{W}} \|XW - X\hat{W}\|_F^2.$$

Although effective, this formulation treats every linear layer independently and ignores the interactions among different components of the transformer architecture.

2 Motivation

Within each attention head, the Query, Key, and Value matrices jointly determine the final attention output.

For a single attention head,

$$A(X) = \text{softmax} \left(\frac{XW_Q(XW_K)^T}{\sqrt{d_k}} \right) XW_V.$$

Conventional quantization minimizes the approximation error of each matrix independently,

$$W_Q, W_K, W_V,$$

despite the fact that only their joint computation determines the attention output.

This motivates a new perspective:

Instead of preserving individual linear transformations, preserve the functionality of the complete attention computation.

3 Project Objectives

The project consists of two complementary research directions.

Objective 1: Attention-Aware Joint Quantization

Instead of minimizing

$$\|XW - X\hat{W}\|,$$

students will formulate a quantization objective directly on the attention output.

Specifically,

$$\min_{\hat{W}_Q, \hat{W}_K, \hat{W}_V} \mathcal{L}(A(X), \hat{A}(X)),$$

where

$$\hat{A}(X) = \text{softmax} \left(\frac{X\hat{W}_Q(X\hat{W}_K)^T}{\sqrt{d_k}} \right) X\hat{W}_V.$$

The optimization should jointly determine

$$(\hat{W}_Q, \hat{W}_K, \hat{W}_V)$$

instead of treating them independently.

Students are encouraged to investigate better optimization strategies, alternative loss functions, calibration methods, and efficient approximation algorithms.

Objective 2: Adaptive Mixed-Precision Quantization

Most quantization methods assign the same precision to every layer.

Instead, determine automatically

$$b_i \in \{2, 3, 4, 8, 16\},$$

where b_i denotes the quantization precision assigned to computational block i .
The optimization problem becomes

$$\begin{aligned} \min_{\{\hat{\theta}_i, b_i\}} \quad & \mathcal{L}(\mathcal{F}(X), \hat{\mathcal{F}}(X)) \\ \text{s.t.} \quad & \sum_i C(b_i) \leq B, \end{aligned}$$

where

- $\mathcal{F}$ denotes the transformer,
- $C(b_i)$ represents memory or computational cost,
- B is the available deployment budget.

The objective is to determine both

- the quantized weights,
- the optimal precision for every computational block.

4 Designing an Importance Criterion

A major challenge is determining which blocks should receive higher precision.

Students are encouraged to propose a novel importance metric.

Possible criteria include

1. Reconstruction error

$$\|f_i(X) - \hat{f}_i(X)\|_F.$$

2. Gradient sensitivity

$$\left\| \frac{\partial \mathcal{L}}{\partial \theta_i} \right\|.$$

3. Hessian approximation.
4. Fisher Information.

5. Activation magnitude.
6. Attention entropy.
7. Singular-value spectrum.
8. Calibration-set sensitivity.
9. Influence on downstream task accuracy.
10. Any novel criterion proposed and justified by the student.

Students are encouraged to investigate whether a theoretically motivated criterion provides better performance than heuristic methods.

5 Minimum Requirements

A successful project must include

- R1.** Literature review.
- R2.** Implementation of at least one existing quantization baseline.
- R3.** Implementation of the proposed attention-aware objective.
- R4.** Joint optimization of W_Q , W_K , and W_V .
- R5.** An adaptive precision allocation strategy.
- R6.** Experimental evaluation on at least one transformer model.
- R7.** Comparison with conventional layer-wise quantization.
- R8.** Complete technical report.

6 Suggested Evaluation Metrics

Depending on the selected model, students may report

- Quantization error
- Attention reconstruction error
- Validation accuracy
- Word Error Rate (ASR)
- Perplexity (Language Models)
- BLEU score (Translation)
- Inference latency
- Memory footprint
- Energy consumption (optional)

7 Deliverables

Students must submit

1. Source code.
2. Technical report (6–10 pages).
3. Experimental results.
4. Presentation (15 minutes).
5. Reproducible instructions.

8 Evaluation

Implementation Quality	20%
Attention-Aware Quantization	20%
Adaptive Precision Strategy	20%
Experimental Evaluation	20%
Research Novelty and Analysis	20%

9 Bonus Challenges

Students aiming for full marks are encouraged to explore one or more of the following directions.

1. Generalize the formulation from one attention head to multi-head attention.
2. Include the output projection matrix W_O .
3. Extend the objective to the entire transformer block.
4. Jointly optimize activation and weight quantization.
5. Develop Hessian-aware optimization.
6. Learn mixed precision automatically.
7. Design calibration-free quantization.
8. Minimize inference latency rather than memory.
9. Generalize the framework to arbitrary computational graphs.
10. Compare against multiple state-of-the-art PTQ methods.
11. Derive theoretical guarantees for the proposed optimization.
12. Demonstrate generalization across multiple transformer architectures (e.g., encoder-only, decoder-only, encoder-decoder, speech transformers, or vision transformers).

10 Expected Research Contribution

The purpose of this project is not simply to reproduce an existing quantization method.

Students are expected to investigate new formulations of transformer quantization based on preserving the functionality of computational blocks instead of individual linear layers.

Outstanding projects should propose

- a mathematically justified optimization objective,
- a principled criterion for allocating quantization precision,
- extensive experimental validation,
- and, ideally, a method that outperforms existing post-training quantization approaches under identical computational budgets.

Such a contribution has the potential to form the basis of a workshop paper or a submission to a leading machine learning systems conference.

Good luck, and think like a researcher!